

Master Recherche (MS)²SC 2025–2026
MG3416–Advanced Structural Dynamics and Acoustics

Tutorial class #1

Exercise 1: Coupled oscillators

The exercise is concerned with the two-degrees-of-freedom system of Fig. 1 constituted by two oscillators in series. Let $\hat{h}_j(\omega) = (-m_j\omega^2 + ic_j\omega + k_j)^{-1}$, $j = 1, 2$, and $\lambda(\omega) = ic_2\omega + k_2$.

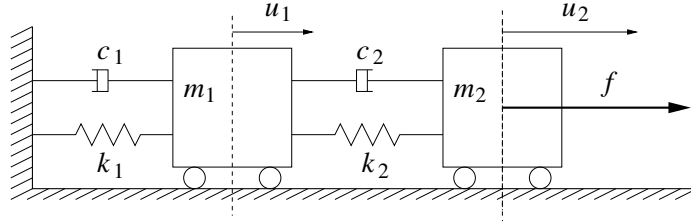


Figure 1: Two-degrees-of-freedom system.

- Q1.** Write the equations of motion of the masses satisfied by u_1 , u_2 , and their time derivatives.
- Q2.** Let $t \mapsto \mathbb{h}_1(t)$ and $t \mapsto \mathbb{h}_2(t)$ be the impulse response functions such that $u_1(t) = \mathbb{h}_1 * f(t)$ and $u_2(t) = \mathbb{h}_2 * f(t)$, where $*$ stands for the time convolution product. Compute $\hat{\mathbb{h}}_1(\omega)$ and $\hat{\mathbb{h}}_2(\omega)$ as functions of $\hat{h}_1(\omega)$, $\hat{h}_2(\omega)$, and $\lambda(\omega)$.
- Q3.** Compute the eigenfrequencies of the coupled system.
- Q4.** The load f is now modeled by the stochastic process $(F_t, t \in \mathbb{R})$ indexed on $\mathbb{R}$, with values in $\mathbb{R}$, which is Gaussian, centered, mean-square stationary and continuous. In addition, its power spectral measure reads $M_F(d\omega) = S_F(\omega)d\omega$. Let $\mathbf{U}_t = (U_{1t}, U_{2t})$ be the stationary response defined by $U_{jt} = \mathbb{h}_j * F_t$, $j = 1, 2$. Compute its power spectral density matrix $\mathbf{S}_U(\omega)$.
- Q5.** Compute $\sigma_1^2 = \mathbb{E}\{U_{1t}^2\}$.
- Q6.** Compute $\tilde{\sigma}_1^2 = \mathbb{E}\{\dot{U}_{1t}^2\}$ in terms of σ_1^2 .
- Q7.** Compute $\mathbb{E}\{U_{1\max}\}$ where $U_{1\max} = \max_{t \in [0, T]} U_{1t}$.

Exercise 2: Random vibrations of a car due to road roughness

The purpose of this example is to study the random vibrations of a car induced by the unevenness of the road. The car is a rigid body of mass M supported by tires modeled by a dashpot with a stiffness K and damping constant D . It is moving at a constant speed $\mathbf{v}_0 = v_0 \mathbf{i}_x$ along the horizontal axis of a fixed reference frame $(0, \mathbf{i}_x, \mathbf{i}_y)$. The road profile is given by the function $x \mapsto y_g(x)$ in this frame. The car vibrations around its static reference configuration will be studied in an attached moving frame $(C(t), \mathbf{e}_x(t), \mathbf{e}_y(t))$ running along the mean flat road. The first part considers the tires without absorbers, and the influence of absorbers between the car body and the tires is considered in a second part.

Car without absorbers

The car has no absorbers and is directly supported by the tires (see Fig. 2).

- Q1.** What are the resonance frequency ω_p and the critical damping rate ζ of the car as functions of K , D and M ?
- Q2.** What is the position vector $\mathbf{x}_T(t)$ of the center of the tire T (equated to the contact point with the road) in the reference frame as function of y_g and v_0 ? What is its position $\tilde{\mathbf{x}}_T(t)$ in the moving frame?

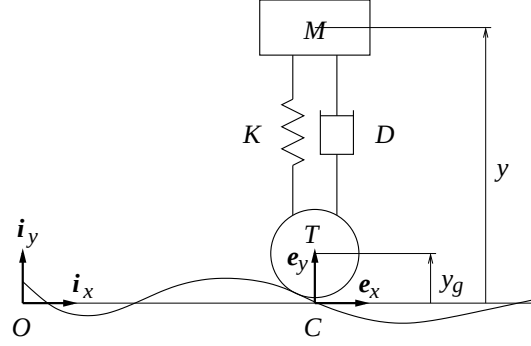


Figure 2: A car on an uneven road; no absorbers.

- Q3.** What is the dynamic equilibrium equation satisfied by the vertical displacement $t \mapsto y(t)$ of the car in the moving frame?
- Q4.** The road profile function $y_g(x)$ is modeled by a second-ordered, mean-zero, mean-square stationary stochastic process $(Y_g(x), x \in \mathbb{R})$. Its auto-correlation function is $x \mapsto R_g(x) = \mathbb{E}\{Y_g(x+x')Y_g(x')\}$ and it is assumed that it has a power spectral density function $k \mapsto S_g(k)$ such that $R_g(x) = \int_{\mathbb{R}} e^{ikx} S_g(k) dk$. What is the random load $(F(t), t \in \mathbb{R})$ applied to the car in the moving frame? Compute its second-ordered characteristics: mean function $m_F(t)$, correlation function $R_F(t)$, and spectral density function $S_F(\omega)$ as functions of R_g , S_g , and v_0 .
- Q5.** Compute the power spectral density function $S_Y(\omega)$ of the stationary response of the car.
- Q6.** Compute the standard deviation σ_Y of the stationary response of the car for the following spectrum of the road profile:

$$S_g(k) = \frac{\lambda^2 k_0}{k^2 + k_0^2},$$

where λ is the standard deviation of the thickness of road roughness and k_0 is a reference spatial frequency.

- Q7.** Assume that $(Y_g(x), x \in \mathbb{R})$ is Gaussian and mean-square derivable. Using S.O. Rice's formula, compute the averaged number of times the car random displacement Y exceeds a threshold y_0 during a period τ and its apparent frequency. Deduce its average maximum over that period. DATA: $\frac{\omega_p}{2\pi} = 5$ Hz, $v_0 = 90$ km/h, $\tau = 1$ h, $y_0 = 5 \times \lambda$; $\lambda = 5$ mm and $k_0 = 100 \text{ m}^{-1}$ (a good road), or $\lambda = 5$ cm and $k_0 = 1 \text{ m}^{-1}$ (a poor road).

Car with absorbers

The car is now supported by tires with absorbers, modeled as a dashpot with a mass μ , stiffness κ and damping δ (see Fig. 3). Influence of the absorbers on the apparent frequency and maximum response.

- Q8.** What are the dynamic equilibrium equations satisfied by the vertical displacements $t \mapsto \psi(t)$ and $t \mapsto y(t)$ of the suspension and the car, respectively, in the moving frame?
- Q9.** What are the new resonance frequencies of this system as functions of $\Omega_1^2 = \frac{K+\kappa}{\mu}$ and $\Omega_2^2 = \frac{\kappa}{M} = \frac{\kappa}{K} \omega_p^2$?
- Q10.** The road profile function $y_g(x)$ is modeled by the same second-ordered, mean-zero, mean-square stationary stochastic process $(Y_g(x), x \in \mathbb{R})$ as for question 4. Compute the spectral density matrix $\mathbf{S}_Z(\omega)$ of the system's stationary random DOFs $\mathbf{Z}(t) = (Y(t), \Psi(t))^T$.
- Q11.** Compute the new standard deviations σ_Y , $\sigma_{\dot{Y}}$ and apparent frequency of the stationary response of the car body for the same road spectrum as for question 6. Make comments on this result.

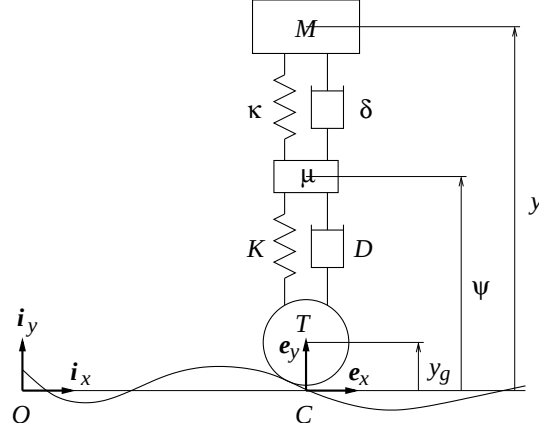


Figure 3: A car on an uneven road; with absorbers.

Important formulas

The solution of the SDOF equation $\ddot{u} + \omega^2 u = f(t)$, $t \geq 0$, with prescribed initial conditions $u(0) = u_0$ and $\dot{u}(0) = v_0$:

$$u(t) = u_0 \cos \omega t + \frac{v_0}{\omega} \sin \omega t + \frac{1}{\omega} \int_0^t \sin \omega(t - \tau) f(\tau) d\tau.$$

An integration formula:¹

$$\int_{-\infty}^{\infty} \left| \frac{b_0 + i\omega b_1 - \omega^2 b_2 - i\omega^3 b_3}{a_0 + i\omega a_1 - \omega^2 a_2 - i\omega^3 a_3 + \omega^4 a_4} \right|^2 d\omega = \pi \frac{a_0 b_3^2 (a_1 a_2 - a_0 a_3) + a_0 a_1 a_4 (b_2^2 - 2b_1 b_3) + a_0 a_3 a_4 (b_1^2 - 2b_0 b_2) + a_4 b_0^2 (a_2 a_3 - a_1 a_4)}{a_0 a_4 (a_1 a_2 a_3 - a_0 a_3^2 - a_1^2 a_4)}, \quad (1)$$

for $a_0, a_1, a_2, a_3, a_4, b_0, b_1, b_2, b_3, b_4 \in \mathbb{R}$.

S.O. Rice formulas – Average rate of u -level upcrossings of a stationary Gaussian process $(X(t), t \in \mathbb{R})$ in an interval I :

$$\mathbb{E}\{U_u(I)\} = \frac{|I|}{2\pi} \frac{\sigma_{\dot{X}}}{\sigma_X} \exp\left(-\frac{u^2}{2\sigma_X^2}\right).$$

Its average maximum $\mathbb{E}\{X_{\max}(I)\}$ in an interval I , $X_{\max}(I) = \sup_{t \in I} X(t)$:

$$\mathbb{E}\{X_{\max}(I)\} = \sigma_X \left(\sqrt{2 \ln(f_X |I|)} + \frac{\Gamma}{\sqrt{2 \ln(f_X |I|)}} \right),$$

where $\Gamma = 0.577\dots$ is Euler's constant and $f_X = \mathbb{E}\{U_0(1)\}$ is the apparent frequency of the process.

¹A. Le Bot, *Foundation of Statistical Energy Analysis in Vibroacoustics*, pp. 291. Oxford University Press, Oxford (2015).

Solutions

Exercise 1

1. The coupled equations of motion are:

$$\begin{aligned} m_1 \ddot{u}_1 + c_1 \dot{u}_1 + k_1 u_1 + c_2(\dot{u}_1 - \dot{u}_2) + k_2(u_1 - u_2) &= 0, \\ m_2 \ddot{u}_2 + c_2(\dot{u}_2 - \dot{u}_1) + k_2(u_2 - u_1) &= f. \end{aligned}$$

2. Taking the Fourier transform of the above equations with the proposed notations yields:

$$\widehat{h}_1(\omega) = \frac{\lambda(\omega) \widehat{h}_1(\omega) \widehat{h}_2(\omega)}{1 + \lambda(\omega) \widehat{h}_1(\omega)(1 - \lambda(\omega) \widehat{h}_2(\omega))}, \quad \widehat{h}_2(\omega) = \frac{(1 + \lambda(\omega) \widehat{h}_1(\omega)) \widehat{h}_2(\omega)}{1 + \lambda(\omega) \widehat{h}_1(\omega)(1 - \lambda(\omega) \widehat{h}_2(\omega))}.$$

3. The eigenfrequencies are:

$$\omega_{\pm}^2 = \frac{1}{2} \left[\frac{k}{m_1} + \frac{k_2}{m_2} \pm \sqrt{\left(\frac{k}{m_1} + \frac{k_2}{m_2} \right)^2 - 4 \frac{k_1 k_2}{m_1 m_2}} \right],$$

with $k = k_1 + k_2$.

4. If $\widehat{h}(\omega) = (\widehat{h}_1(\omega), \widehat{h}_2(\omega))$ then $\mathbf{S}_U(\omega) = S_F(\omega) \widehat{h}(\omega) \otimes \overline{\widehat{h}(\omega)}$. Therefore:

$$\mathbf{S}_U(\omega) = S_F(\omega) \begin{bmatrix} |\widehat{h}_1(\omega)|^2 & \widehat{h}_1(\omega) \overline{\widehat{h}_2(\omega)} \\ \overline{\widehat{h}_1(\omega)} \widehat{h}_2(\omega) & |\widehat{h}_2(\omega)|^2 \end{bmatrix}.$$

5. One has $\sigma_1^2 = \mathbb{E}\{U_{1t}^2\} = R_1(0) = \int [\mathbf{S}_U(\omega)]_{11} d\omega$, that is:

$$\begin{aligned} \sigma_1^2 &= \int \frac{|\lambda(\omega)|^2 |\widehat{h}_1(\omega)|^2 |\widehat{h}_2(\omega)|^2 S_F(\omega) d\omega}{|1 + \lambda(\omega) \widehat{h}_1(\omega) - \lambda(\omega)^2 \widehat{h}_1(\omega) \widehat{h}_2(\omega)|^2} \\ &\simeq S_F(\omega_1) \int \frac{|\lambda(\omega)|^2 d\omega}{|(k_1 + i c_1 \omega - m_1 \omega^2)(\lambda(\omega) - m_2 \omega^2) - m_2 \omega^2 \lambda(\omega)|^2}, \end{aligned}$$

since $1/\widehat{h}_2(\omega) = \lambda(\omega) - m_2 \omega^2$. Here $\omega_1^2 = \frac{k}{m_1}$, the "blocked" eigenfrequency of the first mass. The remaining integral may be computed using the proposed formula with $b_0 = k_2^2$, $b_2 = c_2^2$, $b_4 = 0$, and $a_0 = k_1 k_2$, $a_1 = i(k_1 c_2 + k_2 c_1)$, $a_2 = -(m_1 k_2 + m_2 k + c_1 c_2)$, $a_3 = -i(m_1 c_2 + m_2 c)$, $a_4 = m_1 m_2$, with $c = c_1 + c_2$. The final result is:

$$\sigma_1^2 \simeq \frac{k_2^2 + (\frac{m_2}{m_1})^2 \omega_2^2 c^2}{\pi k c} S_F(\omega_1)$$

with $\omega_2^2 = \frac{k_2}{m_2}$, the "blocked" eigenfrequency of the second mass.

6. Now $\tilde{\sigma}_1^2 = \mathbb{E}\{\dot{U}_{1t}^2\} = \int \omega^2 [\mathbf{S}_U(\omega)]_{11} d\omega$, that is $\tilde{\sigma}_1^2 \simeq \omega_1^2 \sigma_1^2$.

7. It suffices to apply S.O. Rice's formula.

Exercise 2

1. $\omega_p = \sqrt{\frac{K}{M}}$ and $\zeta = \frac{D}{2\sqrt{KM}}$.
2. $\mathbf{x}_T(t) = v_0 t \mathbf{i}_x + y_g(v_0 t) \mathbf{i}_y$, $\tilde{\mathbf{x}}_T(t) = y_g(v_0 t) \mathbf{e}_y$.
3. $M\ddot{y} + D(\dot{y} - \dot{r}) + K(y - r) = 0$, or $M\ddot{y} + D\dot{y} + Ky = D\dot{r} + Kr$ where $r(t) = y_g(v_0 t)$.
4. $F(t) = Dv_0 \dot{Y}_g(v_0 t) + KY_g(v_0 t)$, $m_F(t) = 0$, $R_F(t) = K^2 R_g(v_0 t) - D^2 v_0^2 \ddot{R}_g(v_0 t)$ since:

$$\mathbb{E}\{\dot{Y}_g(x + x') Y_g(x')\} = \frac{dR_g}{dx}(x) = \dot{R}_g(x) = -\mathbb{E}\{Y_g(x + x') \dot{Y}_g(x')\},$$

and:

$$\mathbb{E}\{\dot{Y}_g(x + x') \dot{Y}_g(x')\} = -\frac{d^2 R_g}{dx^2}(x) = -\ddot{R}_g(x).$$

Consequently $S_F(\omega) = \frac{1}{v_0} (K^2 + D^2 \omega^2) S_g(\frac{\omega}{v_0})$ ($k \mapsto S_g(k)$ is necessarily an even function).

5. $S_Y(\omega) = |\hat{h}(\omega)|^2 S_F(\omega)$, where $\hat{h}(\omega) = M^{-1}(\omega_p^2 - \omega^2 + 2i\zeta\omega_p\omega)^{-1}$. This yields:

$$S_Y(\omega) = \frac{1}{v_0} \left(\frac{\omega_p^4 + 4\zeta^2\omega_p^2\omega^2}{(\omega_p^2 - \omega^2)^2 + 4\zeta^2\omega_p^2\omega^2} \right) S_g \left(\frac{\omega}{v_0} \right).$$

6. $\sigma_Y^2(t) = \mathbb{E}\{(Y(t) - m_Y(t))^2\}$ by definition of the standard deviation $t \mapsto \sigma_Y(t)$, that is $\sigma_Y^2 = R_Y(0)$ is a constant whenever $m_Y(t) = 0$ and $(Y(t), t \in \mathbb{R})$ is mean-square stationary. Thus:

$$\begin{aligned} \sigma_Y^2 &= \int_{\mathbb{R}} S_Y(\omega) d\omega = \frac{1}{v_0} \int_{\mathbb{R}} \frac{\omega_p^4 + 4\zeta^2\omega_p^2\omega^2}{(\omega_p^2 - \omega^2)^2 + 4\zeta^2\omega_p^2\omega^2} S_g \left(\frac{\omega}{v_0} \right) d\omega \\ &= \omega_0 \lambda^2 \int_{\mathbb{R}} \frac{(\omega_p^4 + 4\zeta^2\omega_p^2\omega^2) d\omega}{(\omega^2 + \omega_0^2)((\omega_p^2 - \omega^2)^2 + 4\zeta^2\omega_p^2\omega^2)} \\ &= \frac{\pi \lambda^2 \omega_p}{2\zeta \omega_0} \left(\frac{(1 + 4\zeta^2)\omega_0^2 - 2\zeta\omega_p\omega_0}{\omega_p^2 + \omega_0^2 - 2\zeta\omega_p\omega_0} \right), \end{aligned}$$

using the integration formula (1) with $b_0 = \omega_p^4$, $b_2 = 4\zeta^2\omega_p^2$, $b_4 = 0$, $a_0 = i\omega_0\omega_p^2$, $a_1 = \omega_p^2 - 2\zeta\omega_p\omega_0$, $a_2 = i(2\zeta\omega_p - \omega_0)$, $a_3 = -1$, $a_4 = 0$, and $\omega_0 = v_0 k_0$.

7. $\mathbb{E}\{U(\tau, y_0)\} = \frac{\tau}{2\pi} \frac{\sigma_Y}{\sigma_Y} \exp(-\frac{y_0^2}{2\sigma_Y^2})$, and it remains to compute σ_Y :

$$\begin{aligned} \sigma_Y^2 &= \int_{\mathbb{R}} \omega^2 S_Y(\omega) d\omega = \omega_0 \lambda^2 \int_{\mathbb{R}} \frac{(\omega_p^4 + 4\zeta^2\omega_p^2\omega^2)\omega^2 d\omega}{(\omega^2 + \omega_0^2)((\omega_p^2 - \omega^2)^2 + 4\zeta^2\omega_p^2\omega^2)} \\ &= \frac{\pi \lambda^2}{2\zeta} \omega_p \omega_0 \left(\frac{(1 + 4\zeta^2)\omega_p^2 - 8\zeta^3\omega_p\omega_0}{\omega_p^2 + \omega_0^2 - 2\zeta\omega_p\omega_0} \right), \end{aligned}$$

using again the integration formula (1) with the same a_j 's and $b_0 = 0$, $b_2 = \omega_p^4$, $b_4 = 4\zeta^2\omega_p^2$. The apparent frequency of the stationary car response is:

$$f_Y = \frac{1}{2\pi} \frac{\sigma_Y}{\sigma_Y} = \frac{\omega_p}{2\pi} \sqrt{\frac{1 + 4\zeta^2 - 8\zeta^3 \frac{\omega_p}{\omega_0}}{1 + 4\zeta^2 - 2\zeta \frac{\omega_p}{\omega_0}}} \simeq \frac{\omega_p}{2\pi}, \quad \zeta \ll 1.$$

The tire does not filtrate anything from the road unevenness!

8. $M\ddot{y} + \delta(\dot{y} - \dot{\psi}) + \kappa(y - \psi) = 0$ for the car, and $\mu\ddot{\psi} + D(\dot{\psi} - \dot{r}) + K(\psi - r) = \delta(\dot{y} - \dot{\psi}) + \kappa(y - \psi)$ for the suspension. The system equations read for the DOFs ψ and y :

$$\begin{bmatrix} M & 0 \\ 0 & \mu \end{bmatrix} \begin{pmatrix} \ddot{y} \\ \ddot{\psi} \end{pmatrix} + \begin{bmatrix} \delta & -\delta \\ -\delta & D + \delta \end{bmatrix} \begin{pmatrix} \dot{y} \\ \dot{\psi} \end{pmatrix} + \begin{bmatrix} \kappa & -\kappa \\ -\kappa & K + \kappa \end{bmatrix} \begin{pmatrix} y \\ \psi \end{pmatrix} = \begin{pmatrix} 0 \\ D\dot{r} + Kr \end{pmatrix},$$

where $r(t) = y_g(v_0 t)$.

9. $\omega_{\pm}^2 = \frac{1}{2} \left[\Omega_1^2 + \Omega_2^2 \pm \sqrt{(\Omega_1^2 + \Omega_2^2)^2 - 4 \frac{K}{K+\kappa} \Omega_1^2 \Omega_2^2} \right]$.

10. $\mathbf{S}_Z(\omega) = \hat{\mathbf{H}}(\omega) \mathbf{S}_F(\omega) \hat{\mathbf{H}}(\omega)^*$ where:

$$\begin{aligned} \mathbf{S}_F(\omega) &= \begin{bmatrix} 0 & 0 \\ 0 & S_F(\omega) \end{bmatrix}, \\ \hat{\mathbf{H}}(\omega) &= \begin{bmatrix} -M\omega^2 + i\delta\omega + \kappa & -(i\delta\omega + \kappa) \\ -(i\delta\omega + \kappa) & -\mu\omega^2 + i(D + \delta)\omega + K + \kappa \end{bmatrix}^{-1}, \end{aligned}$$

and $S_F(\omega)$ has been computed in question 4; thus:

$$\mathbf{S}_Z(\omega) = S_F(\omega) \begin{bmatrix} |\hat{H}_{12}(\omega)|^2 & \hat{H}_{12}(\omega) \overline{\hat{H}_{22}(\omega)} \\ \overline{\hat{H}_{12}(\omega)} \hat{H}_{22}(\omega) & |\hat{H}_{22}(\omega)|^2 \end{bmatrix}.$$

Observe that $\mathbf{S}_Z(\omega)$ is Hermitian, as expected. Computing the inverse of $\hat{\mathbf{H}}$ it is found for instance that:

$$\hat{H}_{12}(\omega) = \frac{Z_2(\omega)}{Z_2(\omega)(\mu Z_1(\omega) - M Z_2(\omega)) - \mu \omega^2 Z_1(\omega)}$$

with $Z_1(\omega) = -\omega^2 + i\Delta_1\omega + \Omega_1^2$, $Z_2(\omega) = i\Delta_2\omega + \Omega_2^2$, and $\Delta_1 = \frac{D+\delta}{\mu}$, $\Delta_2 = \frac{\delta}{M}$.

11. $\sigma_Y^2 = \int_{\mathbb{R}} |\hat{H}_{12}(\omega)|^2 S_F(\omega) d\omega$ and $\sigma_{\dot{Y}}^2 = \int_{\mathbb{R}} \omega^2 |\hat{H}_{12}(\omega)|^2 S_F(\omega) d\omega$ from the previous result. The derivation of these integrals is carried on with the assumption $\sigma_Y^2 \simeq S_F(\bar{\omega}) \int_{\mathbb{R}} |\hat{H}_{12}(\omega)|^2 d\omega$ and $\sigma_{\dot{Y}}^2 \simeq S_F(\bar{\omega}) \int_{\mathbb{R}} \omega^2 |\hat{H}_{12}(\omega)|^2 d\omega$, where $\bar{\omega} = \frac{1}{2}(\omega_- + \omega_+)$. σ_Y is derived from Eq. (1) with $b_4 = 0$, $b_2 = \Delta_2^2$, and $b_0 = \Omega_2^4$, while $\sigma_{\dot{Y}}$ is derived with $b_4 = \Delta_2^2$, $b_2 = \Omega_2^4$, and $b_0 = 0$; both use $a_4 = 1$, $a_3 = -i(\Delta_1 + \Delta_2)$, $a_2 = -(\Omega_1^2 + \Omega_2^2 + \Delta_1\Delta_2 - \Delta_2^2\omega_q^2/\Omega_2^2)$, $a_1 = i(\Delta_1\Omega_2^2 + \Delta_2\Omega_1^2 - 2\Delta_2\omega_q^2)$, and $a_0 = \omega_p^2\omega_q^2$ with $\omega_q^2 = \frac{\kappa}{\mu}$ (eigenfrequency of the suspensions). After tedious computations it is found that:

$$f_Y \simeq \frac{\omega_p}{2\pi} \sqrt{\frac{M(\Delta_1 + \Delta_2)\Omega_2^2}{\mu(\Delta_1\Omega_1^2 + \Delta_2\Omega_2^2) + 2M\Delta_2\Omega_2^2}}.$$

Since hopefully $\mu \ll M$, one has:

$$f_Y \simeq \frac{\Omega_2}{2\pi} \sqrt{\frac{K}{K + \kappa}} = \frac{\omega_p}{2\pi} \sqrt{\frac{\kappa}{K + \kappa}} \simeq \frac{K}{\kappa} \omega_- ,$$

independently of the damping factors.